This paper covers Differential Geometry, Barrett O'Neill, with the sub Analysis on Manifold, Munkres.

Chapter 1. Calculus on Euclidean Space.

Def. A real-valued $f: \mathbb{R}^3 \to \mathbb{R}$ is called smooth if:

$$\frac{\partial^k f}{\partial x_1^{k_1} \partial x_2^{k_2} \partial x_3^{k_3}} \text{ exists and continuous for all } k_1, k_2, k_3 \geq 0.$$

And, given two smooth functions f and g, define

$$(f+g)(P) = f(P) + g(P) \quad \text{and} \quad (fg)(P) = f(P) \cdot g(P).$$

Def. A tuple $V_P = (P, V) \in \mathbb{R}^3 \times \mathbb{R}^3$ is called a tangent vector, consisting of the point part P and vector part V.

And, the set $T_P(\mathbb{R}^3) \overset{def}{=} \{(P, V) \mid V \in \mathbb{R}^3\}$ is called the tangent space of $\mathbb{R}^3$ at P.

Def. A mapping $V: \mathbb{R}^3 \to \bigcup_{P \in \mathbb{R}^3} T_P(\mathbb{R}^3) : P \mapsto V_P$ is called vector field on $\mathbb{R}^3$.

Given two vector field V and W, and real-valued f on $\mathbb{R}^3$, define.

$$(V+W)(P) = V(P) + W(P) \quad \text{and} \quad (fV)(P) = f(P)V(P).$$

In particular, the vector fields on $\mathbb{R}^3$ define by

$$U_1(P) = (1,0,0)_P \quad \text{and} \quad U_2(P) = (0,1,0)_P \quad \text{and} \quad U_3(P) = (0,0,1)_P \quad \text{for each } P \in \mathbb{R}^3$$

are called the natural frame field on $\mathbb{R}^3$. $\lrcorner$

Lemma. Given a vector field V on $\mathbb{R}^3$, there are unique real-valued V_1, V_2, V_3 on $\mathbb{R}^3$

$$\text{s.t.} \quad V = V_1 U_1 + V_2 U_2 + V_3 U_3.$$

Particularly, the functions V_1, V_2, V_3 are called the Euclidean coordinate functions of V.

Proof. By definition, for each point $P \in \mathbb{R}^3$, the vector part of $V(P)$ depends on P, so, denote the vector part of $V(P)$ as $(V_1(P), V_2(P), V_3(P))$. Now,

$$V(P) = (V_1(P), V_2(P), V_3(P))_P = V_1(P)U_1(P) + V_2(P)U_2(P) + V_3(P)U_3(P).$$

And, the uniqueness is given by the comparing the entries of tuples.

Def. Let f be a smooth real-valued function on $\mathbb{R}^3$, and V_p be a tangent vector. the limit $V_p[f] \overset{\text{def}}{=} \frac{d}{dt}(f(p+tv))\Big|_{t=0} \left(= \lim_{t\to 0} \frac{f(p+tv)-f(p)}{t}\right)$

is called the directional derivative of f with respect to V_p.

Lemma. Given a tangent vector $V_p = (v_1, v_2, v_3)_p$ on $\mathbb{R}^3$ and smooth function f,

$$V_p[f] = \sum_{i=1}^{3} v_i \cdot \frac{\partial f}{\partial x_i}(p)$$

Proof. Let $p = (p_1, p_2, p_3) \in \mathbb{R}^3$ be given. Then

$$p + tv = (p_1 + tv_1, p_2 + tv_2, p_3 + tv_3)$$

So that $f(p+tv) = f(p_1 + tv_1, p_2 + tv_2, p_3 + tv_3)$

Now, by the chain rule and the fact that $\frac{d}{dt}(p_i + tv_i) = v_i$,

$$V_p[f] = \frac{d}{dt}(f(p+tv))\Big|_{t=0} = \sum_{i=1}^{3} \frac{\partial f}{\partial x_i}(p)\, v_i. \quad \blacksquare$$

Note: In other words,

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{dt} + \frac{\partial f}{\partial x_3} \cdot \frac{dx_3}{dt}$$

Thm. Let f and g be functions on $\mathbb{R}^3$, V_p and W_p be tangent vectors, and $a, b \in \mathbb{R}$.

1) $(aV_p + bW_p)[f] = aV_p[f] + bW_p[f]$
2) $V_p[af + bg] = aV_p[f] + bV_p[g]$
3) $V_p[fg] = V_p[f] \cdot g(p) + f(p) \cdot V_p[g]$.

Proof. These are obtained by the Lemma above.

Corollary. Given two vector fields V and W on $\mathbb{R}^3$ and real-valued f, g, h on $\mathbb{R}^3$,

1) $(fV + gW)[h] = fV[h] + gW[h]$
2) $V[af + bg] = aV[f] + bV[g]$ for all $a, b \in \mathbb{R}$.
3) $V[fg] = V[f] \cdot g + fV[g]$.

Proof.

$$(av_p + bw_p)[f] \overset{\text{Lemma 3}}{=} \sum_{i=1} (av_i + bw_i) \cdot \frac{\partial f}{\partial x_i}(p) = a\sum v_i \frac{\partial f}{\partial x_i}(p) + b\sum w_i \cdot \frac{\partial f}{\partial x_i}(p).$$

Sec 1.4 Curves in $\mathbb{R}^3$. (I is any open interval).

Def. A differentiable function $\alpha : I \to \mathbb{R}^3$ is called a curve.
Moreover, for $\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$, the tangent vector

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}$$

is called the velocity vector of α at t.

Examples for curves and the velocity vector.

Reparametrization

Def. Let $\alpha: I \to \mathbb{R}^3$ be a curve defined on an open interval I.
If $h: J \to I$ is a differential function on an open interval J,
then the composite $\beta = \alpha \circ h: J \to \mathbb{R}^3$ is a curve called a reparametrization of α by h.

Lemma. If β is the reparametrization of α by h, then the velocity is

$$\beta'(s) = \frac{dh}{ds}(s) \cdot \alpha'(h(s))$$

Proof. Denote $\alpha = (\alpha_1, \alpha_2, \alpha_3)$. Then, by the definition,

$$\beta(s) = (\alpha \circ h)(s) = (\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s)))$$

Now, by the definition of velocity and the chain rule,

$$\begin{aligned}\beta'(s) &= (\alpha_1'(h(s)) \cdot h'(s), \alpha_2'(h(s)) \cdot h'(s), \alpha_3'(h(s)) \cdot h'(s)) \\ &= h(s) \cdot (\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s))). \\ &= h(s) \cdot \alpha'(h(s)).\end{aligned}$$

Lemma. Let α be a curve in $\mathbb{R}^3$ and let f be a differential function on $\mathbb{R}^3$.
Then $\alpha'(t)[f] = \dfrac{d(f \circ \alpha)}{dt}(t)$

Proof.

1-form.

Def. A map $\varphi : \bigcup_{p \in \mathbb{R}} T_p(\mathbb{R}^3) \to \mathbb{R}$ is called a 1-form on $\mathbb{R}^3$ if;

$$\varphi(av_p + bw_p) = a \cdot \varphi(v_p) + b \cdot \varphi(w_p) \text{ for all } a, b \in \mathbb{R} \text{ and } v_p, w_p \in T_p(\mathbb{R}^3), \text{ for each } p.$$

And, for a fixed $p \in \mathbb{R}^3$, the restriction $\varphi_p : T_p(\mathbb{R}^3) \to \mathbb{R} : v_p \mapsto \varphi(v_p)$ is a linear functional.
Moreover, given two 1-forms φ_1, φ_2 and $f, g : \mathbb{R}^3 \to \mathbb{R}$, define

$$(\varphi_1 + \varphi_2)(v_p) \overset{\text{def}}{=} \varphi_1(v_p) + \varphi_2(v_p) \text{ and } (f\varphi)(v_p) = f(p) \cdot \varphi(v_p), \text{ for each } v_p \in \bigcup_{p \in \mathbb{R}} T_p(\mathbb{R}^3)$$

And given two vector fields V, W,

$$\varphi(fV + gW) \overset{\text{def}}{=} f\varphi(V) + g\varphi(W)$$

and $$(f\varphi_1 + g\varphi_2)(V) \overset{\text{def}}{=} f\varphi_1(V) + g\varphi_2(V).$$

Def. If $f : \mathbb{R}^3 \to \mathbb{R}$ is differentiable, the 1-form

$$df : \bigcup_{p \in \mathbb{R}} T_p(\mathbb{R}^3) \to \mathbb{R} ; v_p \mapsto v_p[f]$$

is called the differential of f.

Example Consider the projections $x_i : \mathbb{R}^3 \to \mathbb{R} : (x_i) \mapsto x_i$ $(i = 1, 2, 3)$. Then,

$$dx_i(v_p) = v_p[x_i] = \sum_{j=1}^{3} v_j \cdot \frac{\partial x_i}{\partial x_j}(p) = v_i.$$

Lemma. Let φ be a 1-form on $\mathbb{R}^3$. Then,

$$\varphi = \sum f_i \, dx_i \quad \text{where } f_i = \varphi(U_i)$$

Proof. For fixed $p \in \mathbb{R}^3$, $\varphi_p = \sum \varphi_p \circ U_p \cdot (dx_i)_p$.

Note: for fixed $p \in \mathbb{R}^3$, the tangent space $T_p(\mathbb{R}^3)$ is isomorphic to $\mathbb{R}^3$.
So, the dual sp. $(T_p(\mathbb{R}^3))^*$ is a set of 1-forms at a. Moreover,

the triple $\{(dx_1)_p, (dx_2)_p, (dx_3)_p\}$ is a dual basis for $(T_p(\mathbb{R}^3))^*$, corresponding to $\{U_1(p), U_2(p), U_3(p)\}$.

「Chapter 2. Frame field.

「Def. Given tangent vectors V_p, W_p at $p \in \mathbb{R}^3$, a tangent vector

$$V_p \times W_p \overset{\text{def}}{=} \begin{vmatrix} U_1(p) & U_2(p) & U_3(p) \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} \overset{(*)}{=} U_1(p)(V_2W_3 - V_3W_2) - U_2(p)(V_1W_3 - V_3W_1) + U_3(p)(V_1W_2 - V_2W_1).$$

is called the cross product of V_p and W_p.

「Lemma. Given V_p and W_p in $T_p(\mathbb{R}^3)$, $V_p \times W_p$ is orthogonal to both V_p and W_p, and

$$\|V_p \times W_p\|^2 = (V_p \cdot V_p)\cdot(W_p \cdot W_p) - (V_p \cdot W_p)^2.$$

In other say, $\|V \times W\| = \|V\| \cdot \|W\| \cdot \sin\theta$

Proof. Since U_1, U_2, U_3 are orthonormal,

$$V \cdot (V \times W) = \begin{vmatrix} V_1 & V_2 & V_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0 \text{ and } W \cdot (V \times W) = \begin{vmatrix} W_1 & W_2 & W_3 \\ V_1 & V_2 & V_3 \\ W_1 & W_2 & W_3 \end{vmatrix} = 0.$$

Therefore, the orthogonality is proved. And, to show the second assertion,

$$(V\cdot V)\cdot(W\cdot W) - (V\cdot W)^2 = (V_1^2+V_2^2+V_3^2)(W_1^2+W_2^2+W_3^2) - (V_1W_1 + V_2W_2 + V_3W_3)^2$$

$$= \sum_{1\le i,j\le 3} V_i^2 \cdot W_j^2 - \Big(\sum_{1\le i\le j\le 3} V_iW_iV_jW_j\Big)$$

$$= \sum_{1\le i,j\le 3} V_i^2 \cdot W_j^2 - \Big(\sum_{i=1}^{3} V_i^2W_i^2 + 2\cdot\sum_{1\le i<j\le 3} V_iW_iV_jW_j\Big)$$

$$= \sum_{1\le i<j\le 3} V_i^2 \cdot W_j^2 - 2\cdot\sum_{1\le i<j\le 3} V_iW_iV_jW_j$$

Meanwhile,

$$\begin{aligned} \|V\times W\|^2 &= (V_2W_3 - V_3W_2)^2 + (V_1W_3 - V_3W_1)^2 + (V_1W_2 - V_2W_1)^2 \\ &= V_2^2W_3^2 + V_3^2W_2^2 - 2V_2W_2V_3W_3 \\ &\quad + V_1^2W_3^2 + V_3^2W_1^2 - 2V_1W_1V_3W_3 \\ &\quad + V_1^2W_2^2 + V_2^2W_1^2 - 2V_1W_1V_2W_2 \end{aligned}$$

These are same. ☒

Sec 2.2 Curves.

Recall: a differentiable function $\alpha : I \to \mathbb{R}^3$ is called a curve on an open interval I.

Def. Given a curve $\alpha : I \to \mathbb{R}^3$,

$$\alpha'(t) \overset{\text{def}}{=} \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)}$$

is called the velocity of α, and $\|\alpha'(t)\|$ is called the speed of α.

And, for $[a,b] \subset I$, $\displaystyle\int_a^b \|\alpha'(t)\|\, dt$

is called the arc length of α from $t=a$ to $t=b$.

Def. A curve $\alpha : I \to \mathbb{R}^3$ is called regular if:

$$\alpha'(t) \neq 0 \text{ for all } t \in I.$$

Thm. Given a regular curve α in $\mathbb{R}^3$,
there exists a reparametrization β of α such that $\|\beta'(s)\| = 1$ for all s.

Proof. Let $\alpha : I \to \mathbb{R}^3$ be a regular curve. Let $a \in I$ be fixed.
Define $S(t) = \displaystyle\int_a^t \|\alpha'(u)\|\, du$. Since $\|\alpha'\| \neq 0$, the $S(t)$ is monotone increasing on I.

Therefore, there is an inverse function $t : J \to I$ s.t. $S(t(x)) = x$ for all $x \in J$,
where $J = S[I]$. Now, for $\beta = \alpha \circ t$,

$$\beta'(s) = \frac{dt}{ds}(s) \cdot \alpha'(t(s)) \text{ and } \|\beta'(s)\| = \underbrace{\frac{dt}{ds}}_{>0}(s) \|\alpha'(t(s))\| = \frac{dt}{ds}(s) \cdot \frac{ds}{dt}(t(s)) = 1.$$

∎

Def. Given a curve $\alpha : I \to \mathbb{R}$, the function

$$Y : I \to \bigcup_{p \in \mathbb{R}^3} T_p(\mathbb{R}^3) : t \mapsto V_{\alpha(t)}$$

is called a vector field on curve α.

「Sec 2.3 Frenet Formula.

「Def. Let $\beta : I \to \mathbb{R}$ be a unit-speed curve.
A vector field β' is called the unit tangent vector field on β, denote $T=\beta'$.
The derivative $T'=\beta''$ is called the curvature vector field of β.
The value $\kappa(s)=\|T'(s)\|$ $(s\in I)$ is called the curvature function of β.」

To restriction s.t $\kappa>0$,
the unit-vector field $N=\frac{T'}{\kappa}$ is called the principal normal vector field of β.
The cross product $B=T\times N$ on β is called the binormal vector field of β.

「Lemma. Let β be a unit-speed curve in $\mathbb{R}^3$ with $\kappa>0$,
Then, the three vector fields $T, N,$ and B on β are unit vector fields that are mutually orthogonal at each point.
Proof. By definition, $\|T\|=\|\beta'\|=1$. And since the curvature $\kappa>0$, $\|N\|=\frac{1}{\kappa}\cdot\|T'\|=1$.
And, since $\|T\|^2=T\cdot T$ is constant, $2T\cdot T'=0$, so $T\perp N$.
therefore the binormal $\|B\|=\|T\times N\|^2=\|T\|^2\cdot\|N\|^2-(T\cdot N)^2=1-0=1$. ∎

In summary, $T=\beta'$, $N=T'/\kappa$, $B=T\times N$, and $\{T, N, B\}$ is orthonormal.
Since $B\cdot T=0$, $B'\cdot T+B\cdot T'=0$ by differentiate. Therefore,

$$B'\cdot T=-B\cdot T'=-B\cdot\kappa N=0$$

And since B is unit, $B'\cdot B=0$, thus, there is a scalar value function τ s.t

$$B'=-\tau N$$

, called the torsion. Now

「Thm. Let β be a unit-speed curve in $\mathbb{R}^3$ with curvature $\kappa>0$ and torsion τ,

$$\begin{aligned} T' &= \qquad \kappa N \\ N' &= -\kappa T \qquad\quad + \tau B \\ B' &= \qquad -\tau N \end{aligned} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{bmatrix}\begin{bmatrix} T \\ N \\ B \end{bmatrix}$$

Proof.

「Curve.

「Def. A differential function $\alpha : (a,b) \to \mathbb{R}^3$ is called a curve.
And, the tangent vector

$$\alpha'(t) \overset{\text{def}}{=} \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t) \right)_{\alpha(t)} \overset{(*)}{=} \sum_{i=1}^{3} \frac{d\alpha_i}{dt}(t) \cdot e_i(\alpha(t)).$$

is called the velocity vector of α at t.
If $h : (c,d) \to (a,b)$ is a differentiable, then the composition

$$\beta = \alpha \circ h : J \to \mathbb{R}^3$$

is called a reparametrization of α by h.

「Lemma. If β is the reparametrization of α by h, then

$$\beta'(s) = h'(s) \cdot \alpha'(h(s)).$$

Chapter 3. Euclidean Geometry.

Preliminary: Mappings.

Def. Given a function $F: \mathbb{R}^n \to \mathbb{R}^m$, define $f_i = \pi_i \circ F$ for each $1 \le i \le m$. Then

$$F(p) = (f_1(p), \cdots, f_m(p)) \quad \text{for all } p \in \mathbb{R}^n$$

these $f_1, \cdots, f_m$ are called the Euclidean Coordinate functions of F.
If the Coordinate functions are differentiable, then F is a mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Def. Given a Curve $\alpha: I \to \mathbb{R}^n$ and mapping $F: \mathbb{R}^n \to \mathbb{R}^m$, a Composition

$$\beta : F \circ \alpha : I \to \mathbb{R}^m$$

is called the image of α under F.

Def. Let $F: \mathbb{R}^n \to \mathbb{R}^m$ be a mapping define

$$F_* : \bigcup_{p \in \mathbb{R}^n} T_p \mathbb{R}^n \to \bigcup_{p \in \mathbb{R}^n} T_{F(p)} \mathbb{R}^m : V_p \mapsto \beta'(0)$$

where $\beta : (-\varepsilon, \varepsilon) \to \mathbb{R}^m : t \mapsto F(p + tv)$.

Prop. Let $F = (f_1, \cdots, f_m)$ be a mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$.
Given a tangent vector $V_p \in T_p(\mathbb{R}^n)$,

$$F_*(V_p) = (V_p[f_1], \cdots, V_p[f_m])_{F(p)}$$

Proof. Since $\beta(t) = (f_1(p + tv), \cdots, f_m(p + tv))$,

$$\frac{d}{dt} f_i(p + tv)\Big|_{t=0} = V_p[f_i],$$ So proved directly by the definition. ☒

Note: for fixed $p \in \mathbb{R}^n$, $F_{*p} : T_p(\mathbb{R}^n) \to T_{F(p)}(\mathbb{R}^m)$ is called the tangent map of F at p.
Moreover, the F_{*p} is linear map.

「Sec 3.2. The tangent map of an Isometry.

「Thm. Let F be an isometry of $\mathbb{R}^3$ with decomposition $F = TC$. Then

$$F_*(v_p) = (C \cdot v)_{F(p)}$$

Chapter 4. Calculus on a Surface.

Def. Let $D \subseteq \mathbb{R}^2$ be an open set.

A one-to-one regular mapping $X : D \to \mathbb{R}^3$ is called a Coordinate Patch.

X_{*p} is 1-1 for each p.

A subset $M \subseteq \mathbb{R}^3$ is called a Surface if:

for each $p \in M$, there exists a patch X such that $N \subseteq \operatorname{Im} X$,

where N is a neighborhood of p in M.

Example. Let $S^2 = \{ (p_1, p_2, p_3) \in \mathbb{R}^3 \mid \sqrt{p_1^2 + p_2^2 + p_3^2} = 1 \}$ be given.

Find a proper patch in S^2 covering a neiborhood of the north Pole $(0,0,1)$.

Let $D = \{ (u,v) \mid u^2 + v^2 < 1 \}$, and define

$$X : D \to \mathbb{R}^3 ; (u,v) \mapsto (u, v, \underbrace{\sqrt{1-u^2-v^2}}_{= f})$$

To show this X satisfies the Patch Condition, calculate the Jacobian

$$J(X) = \begin{pmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial v} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial v} \\ \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ \frac{-u}{f} & \frac{-v}{f} \end{pmatrix}$$

Trivially rank $J = 2$. So, by the inverse function theorem, X_{*p} is locally one-to-one.

And, $X^{-1} : X[D] \to D ; (p_1, p_2, p_3) \mapsto (p_1, p_2)$ is an inverse of X, So one-to-one.

In special case; Given differentiable $f : \mathbb{R}^2 \to \mathbb{R}$, $M = \{ (x, y, f(x,y)) \mid x, y \in \mathbb{R}^2 \}$ is a surface.

Via the Monge patch $X(u,v) = (u, v, f(u,v))$.

Patch Computation.

Def. Given a patch $X: D \to \mathbb{R}^3$, for fixed $(u_0, v_0) \in D$, define

$$u \mapsto X(u, v_0) \quad \text{and} \quad v \mapsto X(u_0, v)$$

are called u, v - parameter curves, respectively.
And, partial velocities of X at (u_0, v_0) as

$$X_u(u_0, v_0) \overset{def}{=} \left(\frac{\partial X}{\partial u}(u_0, v_0)\right)_{(u_0, v_0)} \quad \text{and} \quad X_v(u_0, v_0) \overset{def}{=} \left(\frac{\partial X}{\partial v}(u_0, v_0)\right)_{(u_0, v_0)}$$

Def. A regular mapping $X: D \to \mathbb{R}^3$ s.t. $X[D] \subseteq M$ for the surface M is called a parametrization of the region $X[D]$ in M.

Thm. Given a mapping $X: D \to \mathbb{R}^3$ where $D \subseteq \mathbb{R}^2$ is open,
X is regular if and only if the cross product $X_u \times X_v \neq 0$ for all $p \in D$.

Proof. By definition, X is regular if and only if given $p \in D$, the Jacobian
$DX(p)$ is one-to-one, that is, $\ker DX(p) = \{0\}$, or $\operatorname{rank} DX(p) = 2$.

「The geographical patch in the sphere.

Let $\Sigma \overset{\text{def}}{=} \{(x,y,z) \in \mathbb{R}^3 \mid \sqrt{x^2+y^2+z^2} = r\}$ where $r>0$.

Define a patch

$$X : (-\pi, \pi) \times (-\tfrac{\pi}{2}, \tfrac{\pi}{2}) : (u,v) \mapsto (r\cdot\cos u \cos v,\ r\cdot\sin u\cdot\cos v,\ r\cdot\sin v)$$

$$(= r\cdot\cos v\cdot(\cos u, \sin u, 0) + r\cdot\sin v\cdot(0,0,1).$$

$$(= r\cdot\cos v\cdot\cos u\cdot e_1 + r\cdot\cos v\cdot\sin u\cdot e_2 + r\cdot\sin v\cdot e_3).$$

Calculate the partial velocities

$$X_u(u,v) = (-r\cdot\sin u\cdot\cos v,\ r\cdot\cos u\cdot\cos v,\ 0)$$

$$X_v(u,v) = (-r\cdot\cos u\cdot\sin v,\ -r\cdot\sin u\cdot\sin v,\ r\cdot\cos v).$$

「Surface Criterion.

Let $g: \mathbb{R}^3 \to \mathbb{R}$ be differentiable, and $c \in \mathbb{R}$, and let $M = \{(x, y, z) \in \mathbb{R}^3 \mid g(x, y, z) = c\}$.
If the differential dg is not zero for all $p \in M$, then M is a surface.